



# Detecting Sarcasm in Code-Switched Turkish–English Fan Discourse

ARI 510 ORIGINAL PROJECT PROPOSAL · FALL 2026

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## 1. Team Member and Course Section

- Chaima Zaghouani — Graduate section (ARI 510)
- Approved to work solo (instructor approval on file for this course)

## 2. Task Title and Overview

### Task Title

Detecting Sarcasm in Code-Switched Turkish–English Fan Discourse

### Overview

This project focuses on detecting sarcasm in Turkish–English code-switched fan discussions about Turkish television dramas. The system takes a parent comment (or video/post caption) and its reply as input and classifies the reply as **Sarcastic, Non-sarcastic, or Ambiguous/Uncertain**. Code-switching occurs when a speaker uses Turkish and English within the same comment.

For example:

**Parent:** “He lied to her again.”

**Reply:** “Ne kadar dürüst bir adam, **really** amazing 😊.”

**Label:** Sarcastic

The reply mixes Turkish and English. Its literal meaning sounds positive (“What an honest man, really amazing”), but the parent comment shows that the speaker actually means the opposite. Therefore, the system needs both the **parent comment** and the **reply** to correctly understand the sarcasm.

## Significance

Sarcasm detection is a challenging NLP task because the intended meaning can be different from the literal meaning of the words. This becomes more complex in code-switched conversations, where speakers move between two languages and may also use slang, emojis, or culturally specific expressions.

This project focuses specifically on Turkish–English code-switched fan discourse. In addition to building a sarcasm classifier, I will study whether the amount of language mixing affects model performance and how consistently human annotators agree when labeling these comments.

## What Makes It Challenging

The main challenge is that sarcasm often depends on context. The same reply can be non-sarcastic or sarcastic depending on the comment it responds to. Therefore, the model must understand the relationship between the parent comment and the reply rather than classify the reply by itself.

Code-switching adds another challenge because a single comment may combine Turkish and English words, expressions, slang, and tone markers. The model must understand both languages and the conversational meaning at the same time. Also, code-switching does not automatically mean that a comment is sarcastic, so the model must learn to distinguish language mixing from sarcasm itself.

## 3. Task Definition

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- **Data type:** Text pairs consisting of a parent comment (or video/post caption) and a reply to it.
- **Task type:** Single-label, multi-class text classification.
- **Classes (3): Sarcastic, Non-sarcastic, and Ambiguous/Uncertain.** The third class is included because some comments may not provide enough evidence to confidently decide whether the reply is sarcastic or non-sarcastic. It also allows me to examine cases where human annotators disagree.

| Label                       | When to use it                                                                                                                      | Simple question to ask                                                                |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sarcastic</b>            | The reply says or implies something different from its literal wording, usually to mock, criticize, exaggerate, or joke ironically. | Does the writer appear to mean something different from what the words literally say? |
| <b>Non-sarcastic</b>        | The reply expresses its meaning directly and sincerely.                                                                             | Does the writer appear to mean what they wrote?                                       |
| <b>Ambiguous/Un certain</b> | There is not enough evidence to confidently choose Sarcastic or Non-sarcastic, even after reading the parent comment.               | Could this reasonably be interpreted either way?                                      |

- **Additional metadata field (for analysis only):** `language_mix` — Turkish-dominant, English-dominant, or Mixed. All accepted replies will contain both Turkish and English. The parent comment provides conversational context and may be English-only, Turkish-only, or code-switched. This field simply describes the degree of language mixing: whether the reply contains mostly Turkish, mostly English, or a more balanced mixture of both languages. It will be used to analyze whether model performance changes with the degree of language mixing.

### Defining “genuine” code-switching

A reply is considered genuinely code-switched only when it contains meaningful Turkish and English words or phrases within the same reply. A comment is **not** considered code-switched simply because it contains a person's name, a show name, a URL, an acronym, or one isolated borrowed word. More detailed inclusion and exclusion rules, including examples of borderline cases, will be defined in the annotation guidelines.

## 4. Proposed Evaluation Metric

### Primary metric: Macro-F1

The primary evaluation metric will be **Macro-F1**. The final dataset may be imbalanced across the three classes: Sarcastic, Non-sarcastic, and Ambiguous/Uncertain. Accuracy alone may therefore give a misleading result if one class appears much more often than the others. Macro-F1 calculates the F1 score for each class separately and gives equal importance to all three classes.

## Secondary metrics

I will also report **accuracy, per-class F1, and a confusion matrix**. Accuracy will show the overall percentage of correct predictions, while per-class F1 will show how well the model performs on each of the three classes. The confusion matrix will show which classes the model tends to confuse.

I will also compare model performance across the three **language\_mix** categories: Turkish-dominant, English-dominant, and Mixed to examine whether the degree of language mixing affects performance.

**Inter-annotator agreement** will be measured using Cohen's kappa or Fleiss' kappa, depending on the number of annotators per item. This will show how consistently human annotators assign the three labels, especially for cases labeled Ambiguous/Uncertain.

## 5. Why This Task Is Non-Trivial for a Frontier LLM

Sarcasm can be difficult for a frontier LLM because the intended meaning may depend on conversational context, language mixing, slang, emojis, and culturally specific expressions. This project combines these challenges by focusing on Turkish–English code-switched fan discussions.

### Context Dependency

The meaning of a reply can change depending on the comment it responds to. A reply that appears sincere when read alone may become sarcastic when the parent comment is provided.

For example:

**Parent:** “He lied to her again.”

**Reply:** “Ne kadar dürüst bir adam, really amazing 😊.”

When read literally, the reply describes the character as honest and amazing. However, the parent comment shows that the intended meaning is the opposite. Therefore, the relationship between the parent comment and the reply is important for classification.

### Code-Switching

Code-switching adds another challenge because a single reply may combine Turkish and English words, expressions, slang, and tone markers. The model must interpret both languages

while also determining the intended tone. Some expressions may also depend on cultural or conversational context rather than their literal meaning.

### **Frontier LLM Reality Check**

To test whether this task is already easy for a frontier LLM, I evaluated 10 representative constructed Turkish–English code-switched parent–reply pairs using Claude. For each example, Claude first classified the reply without the parent context and then classified the same reply with the parent context. Both predictions were compared with the human-assigned label: Sarcastic, Non-sarcastic, or Ambiguous/Uncertain.

This comparison shows whether conversational context changes the model's prediction and identifies cases where the model remains incorrect or uncertain even when the parent context is provided.

| # | Parent Comment + Reply                                                                                                                                                              | Human Label   | LLM: Reply Only | LLM: With Context | Observation                                                                                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------|-------------------|---------------------------------------------------------------------------------------------------------------------|
| 1 | <p><b>Parent:</b> “I loved their scene together. They finally talked honestly about everything.”</p> <p><b>Reply:</b> “Kesinlikle, their chemistry was gerçekten beautiful ❤️.”</p> | Non-sarcastic | Non-sarcastic   | Non-sarcastic     | <b>Agreement.</b> The model correctly identifies the sincere praise both with and without context.                  |
| 2 | <p><b>Parent:</b> “That scene made me cry. I wasn't expecting him to leave.”</p> <p><b>Reply:</b> “Ben de ağladım, that scene was gerçekten heartbreaking 😭.”</p>                   | Non-sarcastic | Non-sarcastic   | Non-sarcastic     | <b>Agreement.</b> The model correctly interprets the strong emotional language as non-sarcastic in both conditions. |
| 3 | <p><b>Parent:</b> “She finally stood up for herself and left him.” <b>Reply:</b> “Sonunda! I'm actually çok proud of her.”</p>                                                      | Non-sarcastic | Non-sarcastic   | Non-sarcastic     | <b>Agreement.</b> The model correctly recognizes the positive reaction as sincere in both conditions.               |

|   |                                                                                                                                                                           |           |                      |                      |                                                                                                                                                 |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p><b>Parent:</b> “He promised he would finally tell her everything, but then he lied to her again.”</p> <p><b>Reply:</b> “Gerçekten çok dürüst, I'm impressed.”</p>      | Sarcastic | Ambiguous /Uncertain | Sarcastic            | <b>Context helped.</b> The model is uncertain when reading the reply alone but correctly identifies sarcasm after receiving the parent comment. |
| 5 | <p><b>Parent:</b> “She said she would never forgive him, then forgave him five minutes later.”</p> <p><b>Reply:</b> “Karakter gelişimi gerçekten amazing.”</p>            | Sarcastic | Non-sarcastic        | Ambiguous /Uncertain | <b>Still challenging.</b> The model predicts Non-sarcastic without context and remains uncertain even after receiving the parent comment.       |
| 6 | <p><b>Parent:</b> “He disappeared for three years and came back expecting everyone to trust him immediately.”</p> <p><b>Reply:</b> “Bence he handled that perfectly.”</p> | Sarcastic | Non-sarcastic        | Sarcastic            | <b>Context helped.</b> The prediction changes from Non-sarcastic to Sarcastic after the parent comment is provided.                             |

|   |                                                                                                                                                                                |                      |                      |                     |                                                                                                                                                          |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 | <p><b>Parent:</b> "She hid the truth for the entire season and finally told him in the last episode."</p> <p><b>Reply:</b> "Sonunda, that was quick."</p>                      | Sarcastic            | Ambiguous /Uncertain | Sarcastic           | <b>Context helped.</b> The model is uncertain from the reply alone but correctly identifies sarcasm with the parent context.                             |
| 8 | <p><b>Parent:</b> "He caused the whole problem and now everyone is thanking him for fixing it."</p> <p><b>Reply:</b> "Gerçekten kahraman, what would they do without him."</p> | Sarcastic            | Ambiguous /Uncertain | Sarcastic           | <b>Context helped.</b> The model is uncertain from the reply alone but correctly identifies the ironic praise after receiving the parent.                |
| 9 | <p><b>Parent:</b> "He apologized to her and admitted that everything was his fault."</p> <p><b>Reply:</b> "Vay be, character development 🤔."</p>                               | Ambiguous /Uncertain | Non-sarcastic        | Ambiguous/Uncertain | <b>Context helped.</b> Without context, the model interprets the reply as sincere; with context, it recognizes that the intended tone remains uncertain. |

|    |                                                                                                                                                |                             |           |           |                                                                                                                                                                   |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | <p><b>Parent:</b> “She decided to forgive him after he apologized.”</p> <p><b>Reply:</b> “İyi bakalım, maybe this time will be different.”</p> | Ambiguous<br>/<br>Uncertain | Sarcastic | Sarcastic | <p><b>Disagreement.</b> The model predicts sarcasm both with and without context, while the reply could reasonably express either sincere hope or skepticism.</p> |
|----|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-----------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Preliminary observation.** The 10 constructed examples were tested using Claude under two conditions: reply only and parent comment + reply. The same classification instructions and three labels were used in both conditions. The results show that conversational context helped Claude correctly identify several sarcastic examples that were uncertain or misclassified when the reply was presented alone. However, some cases remained uncertain or disagreed with the human label even after context was provided. These preliminary results suggest that the task is not trivial for a frontier LLM and that parent-comment context can play an important role in sarcasm detection.

## 6. Data Source(s) and Collection Plan

- **Primary source:** YouTube comments from official or major fan-run channels for popular Turkish television dramas. Comments will be collected using the YouTube Data API. YouTube comment threads preserve the parent/reply structure needed to create the input pairs for this task.
- **Secondary source:** Reddit communities focused on Turkish television dramas. Reddit will be used as a supplementary source when relevant Turkish–English code-switched parent/reply pairs are available.
- **Platform requirements:** Data collection will use official platform access methods and will follow applicable platform requirements and rate limits.

### Licensing and Data Release

Source comments will be collected through official platform access methods and handled according to the applicable YouTube and Reddit platform terms. Because the source comments

are user-generated content, I will not assume ownership or relicense the original comment text. For any public dataset release, I will distribute only the fields permitted by the applicable platform requirements, such as comment IDs and project-created annotations when necessary. The annotations and other original metadata created specifically for this project will be released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, while the original comment text will remain subject to the applicable source-platform terms.

### **Code-Switch Candidate Detection**

Raw comments will first be filtered using language-identification methods and simple lexical rules to identify likely Turkish–English code-switched candidates. The flagged comments will then be manually checked using the inclusion and exclusion criteria defined in Section 3. This filtering step only identifies potential code-switched comments; it does **not** assign the sarcasm label.

### **Privacy and Data Handling**

Username and other unnecessary user-identifying information will not be used as model inputs. The working dataset will contain the parent and reply text and only the metadata needed for data organization and annotation, such as comment IDs and timestamps.

Duplicates, spam, advertisements, and comments without sufficient conversational context will be removed during data cleaning.

### **Dataset Sharing**

Because the original comments are user-generated content, the public version of the dataset will avoid redistributing unnecessary user information. Where required by platform rules, the shared dataset will contain comment IDs and the annotations created for this project rather than republishing the complete raw comments. The course instructor will be provided with the data needed to evaluate the project in a form consistent with the applicable platform requirements.

### **Unlabeled-Data Compliance**

The raw comments do not contain sarcasm labels. The labels **Sarcastic, Non-sarcastic, and Ambiguous/Uncertain** will be created during the annotation process rather than taken from an existing labeled dataset. **This ensures that the classification labels are created specifically for this project.**

## 7. Expected Dataset Size and Example Data Point

The initial target is approximately **800–1,200 usable comment pairs**, subject to adjustment based on the yield observed during data collection. I will begin with a **50–100 item pilot batch** to test and refine the annotation guidelines and estimate how many raw comments must be collected to obtain genuinely Turkish–English code-switched pairs.

Many raw comments are expected to be Turkish-only, English-only, spam, duplicates, or otherwise unsuitable. Therefore, the number of raw comments collected will be larger than the final usable dataset. The pilot batch will be used to estimate this ratio before scaling up the collection.

### Example Data Point

| Field          | Value                                            |
|----------------|--------------------------------------------------|
| id             | yt_003821                                        |
| platform       | YouTube                                          |
| parent_comment | Kemal finally told Ayşe the truth this episode 🤔 |
| child_comment  | Vay be, what a shocker, sadece 40 bölüm sürdü 😬. |
| language_mix   | Mixed                                            |
| label          | Sarcastic                                        |

## 8. Solo Work-Plan

### (1) Project Scope and Why It Is Appropriate for a Single Student

This project is designed to be manageable by one student. It uses one clearly defined three-class classification task and a text-only modeling pipeline. The data will be collected through the sources described in Section 6 rather than through custom web scraping.

The initial target of **800–1,200 usable comment pairs** is smaller than a typical team dataset and may be adjusted based on the results of the 50–100 item pilot batch. This allows the amount of data collection and annotation to remain realistic for a solo project.

### (2) Rough Milestone Timeline

- **Weeks 1–2:** Finalize the annotation guidelines, including the code-switching inclusion and exclusion rules. Collect and annotate the 50–100 item pilot batch.
- **Weeks 3–5:** Refine the annotation guidelines based on the pilot results. Continue collecting, cleaning, and annotating the full dataset.

- **Week 6:** Present the task and annotation setup for Checkpoint 1 and begin cross-annotation of other students' datasets.
- **Weeks 7–8:** Compute inter-annotator agreement, review disagreements, and finalize the dataset and its statistics.
- **Weeks 9–10:** Train baseline models, beginning with **TF-IDF + Logistic Regression**, followed by a transformer-based model. Set up the benchmark competition page.
- **Weeks 11–13:** Participate in the required class benchmark competitions and improve the proposed model based on experimental results.
- **Weeks 14–15:** Build and deploy a Streamlit application where users can enter a parent comment and a code-switched reply and receive the model's predicted sarcasm label.

### **(3) Acknowledgment**

I understand that, as an approved solo student, I am responsible for the full project workload, including data collection, annotation guideline development, cross-annotation, benchmark creation, modeling, evaluation, and deployment. The dataset size and project scope are designed accordingly.